INVERTIS UNIVERSITY, BAREILLY

B.Tech CSE – Third Year

APPLIED DATA STRUCTURES & ALGORITHMS LAB (BCS553)

LAB 9

Assignment 9: Constructors

Assignment aim: Initialize object state using constructors.

Experiment No. 9.1 — Default Box constructor

1. Experiment Title: Default Box constructor

2. Aim: Implement the task specified in the BCS553 lab assignment.

3. Software / Tools Required: Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

4. Theory: A constructor runs during object creation and initializes fields; this refers to the current object.

5. Algorithm / Procedure:
Define a no-argument constructor that sets width to 10; create a Box; print width.
Then compile and execute the Java program.

6. Program:

```
class Box {
    int width;
    Box() {
        width = 10;
    }
}
public class Task_01_Default_Box_constructor {
    public static void main(String[] args) {
        Box b = new Box();
        System.out.println("Width: " + b.width);
    }
}
```

7. Input: Fixed input values are declared in the program: Box width or Counter initial value specified in the task.

8. Output:
Width: 10

Time / Space Complexity: O(1) time; O(1) auxiliary space.

9. Result: The program implements the specified task and produces the expected output for the given test data.

10. Viva-Voce Questions:

- What is a constructor?
- When does a constructor run?
- Does a constructor have a return type?

Experiment No. 9.2 — Parameterized Box constructor

1. Experiment Title: Parameterized Box constructor

2. Aim: Implement the task specified in the BCS553 lab assignment.

3. Software / Tools Required: Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

4. Theory: A constructor runs during object creation and initializes fields; this refers to the current object.

5. Algorithm / Procedure:
Pass 25 to the constructor; use this.width to refer to the object's field; print the stored value.
Then compile and execute the Java program.

6. Program:

```
class Box {
    int width;
    Box(int width) {
        this.width = width;
    }
}
public class Task_02_Parameterized_Box_constructor{
    public static void main(String[] args) {
        Box b = new Box(25);
        System.out.println("Width: " + b.width);
    }
}
```

7. Input: Fixed input values are declared in the program: Box width or Counter initial value specified in the task.

8. Output:
Width: 25

Time / Space Complexity: O(1) time; O(1) auxiliary space.

9. Result: The program implements the specified task and produces the expected output for the given test data.

10. Viva-Voce Questions:

- What is a parameterized constructor?
- What does this refer to?
- Why use this.width?

Experiment No. 9.3 — Increment a Counter three times

1. Experiment Title: Increment a Counter three times

2. Aim: Implement the task specified in the BCS553 lab assignment.

3. Software / Tools Required: Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

4. Theory: A constructor runs during object creation and initializes fields; this refers to the current object.

5. Algorithm / Procedure:
Initialize count to zero in the constructor; call increment three times; print count.
Then compile and execute the Java program.

6. Program:

```
class Counter {
    int count;
    Counter() { count = 0; }
    void increment() { count++; }
}
public class Task_03_Increment_a_Counter_three_times {
    public static void main(String[] args) {
        Counter c = new Counter();
        c.increment(); c.increment(); c.increment();
        System.out.println("Count: " + c.count);
    }
}
```

7. Input: Fixed input values are declared in the program: Box width or Counter initial value specified in the task.

8. Output:
Count: 3

Time / Space Complexity: O(1) time; O(1) auxiliary space.

9. Result: The program implements the specified task and produces the expected output for the given test data.

10. Viva-Voce Questions:

- What does count++ do?
- Why initialize count in the constructor?
- What is object state?